Equity Performance Project

Analysis of Investor Personas and Index Tracking

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November 2025

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1 Introduction

Modern portfolio theory (MPT), or mean-variance analysis, is a mathematical framework for assembling a portfolio of assets such that the expected return is maximized for a given level of risk. It is a formalization and extension of diversification in investing, the idea that owning different kinds of financial assets is less risky than owning only one type. Its key insight is that an asset's risk and return should not be assessed by itself, but by how it contributes to a portfolio's overall risk and return. The variance of return (or its transformation, the standard deviation) is used as a measure of risk, because it is tractable when assets are combined into portfolios.

2 Problem and Model Formulation

The main objective of this project is to analyze the performance of a specific equity portfolio from the perspectives of two different hypothetical personas. This analysis will be extended to include a formal evaluation of risk-adjusted returns and the construction of an index-tracking portfolio.

2.1 Core problem definition

The core analysis is centered on a portfolio composed of 10 U.S. stocks, which were selected to represent a mix of technology, dividend aristocrat[1], and high-dividend yield sectors. The portfolio is defined by

- **Technology:** MSFT, AAPL, IBM
- **Dividend aristocrat:** XOM, WTM, KO, JNJ
- **High dividend yield:** VZ, KHC, USB

The performance of this 10-stock target portfolio will be evaluated using historical market data, we assume some specific weighting strategies defined in Section 4.1.

[1]A dividend aristocrat is an S&P 500 company that has increased its dividend annually for at least 25 years, demonstrating a commitment to returning cash to investors

2.2 Persona models (Stochastic Models)

The performance of the target portfolio is evaluated through two personas, each representing a different investment horizon and objective. The P&L for each is modeled as a stochastic process dependent on the historical price path of the portfolio's assets.

2.2.1 Buy & Hold (BH) model

This persona represents a long-term investor who acquires the portfolio at the beginning of the analysis period and holds it until the end.

- **Objective:** Maximize total return over the entire period, inclusive of all cash dividends.
- **Model:** The P&L is a single-stage calculation: $\text{P\&L}_{\text{BH}} = \text{Value}_E - \text{Value}_S$

2.2.2 Hedge Fund (HF) model

This persona represents an institutional investor that marks-to-market[2] and captures P&L changes on a daily basis.

- **Objective:** Generate consistent, positive daily returns (*alpha*) while managing daily volatility.
- **Model:** The PnL is a multi-stage stochastic series: $\text{P\&L}_{HF}(t) = \text{Value}_t - \text{Value}_{t-1}$ for each day t.

2.3 Risk Constraints and Performance Metrics

A simple return analysis is insufficient. We introduce risk constraints by evaluating each persona's P&L stream using standard performance metrics.

- **Sharpe Ratio:** Measures risk-adjusted return, defined as:

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R_p - R_f]}{\sigma[R_p - R_f]} \tag{1}$$

[2] Accounting method that values assets and liabilities at their current market price rather than their historical or original cost

where R_p is the portfolio's return, R_f is the risk-free rate, and σ_p is the standard deviation (volatility) of the portfolio's excess return.

- **Value at Risk (VaR):** Defines the maximum expected loss at a given confidence level $1 - \alpha$. It is the $1 - \alpha$ quantile of the P&L distribution, $F_X(x)$.

$$\mathrm{VaR}_\alpha(X) = -\inf\{x \in \mathbb{R} : F_X(x) > \alpha\} \tag{2}$$

  For this project, we will use the non-parametric Historical Simulation method.

- **Conditional Value at Risk (CVaR):** Measures the expected loss *given* that the loss exceeds the VaR.

$$\mathrm{CVaR}_\alpha(X) = \mathbb{E}[-X \mid -X \geq \mathrm{VaR}_\alpha(X)] \tag{3}$$

2.4 Index Tracking Model (Optimization Model)

This task involves creating a replicating portfolio from a universe of other stocks (e.g., S&P 500 components excluding the 10 target stocks) to mimic the returns of our target portfolio. This is formulated as a constrained optimization problem.

- **Objective Function:** Minimize the Tracking Error Variance (TEV), which is the variance of the difference between the target portfolio's returns (R_T) and the replicating portfolio's returns (R_R).

$$\min_{w} \quad \mathrm{TEV} = E\left[(R_T(t) - R_R(t))^2\right] \tag{4}$$

  where $R_R(t) = \sum_{i=1}^{N} w_i R_i(t)$ and w_i is the weight of stock i in the replicating portfolio.

- **Constraints:** The optimization is subject to several constraints:

  1. **Full Investment:** Weights must sum to one.

$$\sum_{i=1}^{N} w_i = 1 \tag{5}$$

  2. **Long-Only:** No short selling is permitted.

$$w_i \geq 0 \quad \forall i \in \{1, \ldots, N\} \tag{6}$$

  3. **Cardinality Constraint:** The number of stocks in the replicating portfolio must not exceed k (e.g., $k = 20$). This is represented using the L_0-norm:

$$\|w\|_0 \leq k \tag{7}$$

  4. **Buy-In Constraint:** Any selected stock must have a minimum weight of l (e.g., $l = 1\%$). This makes the weights semi-continuous:

$$w_i \in \{0\} \cup [l, 1] \quad \forall i \tag{8}$$

3 Approach and Description of Solution Methods

3.1 Data Acquisition and Preparation

The first stage of the project consisted of collecting historical price data for the selected U.S. equities over a multi-year horizon. Following the project requirements, we used a data window ranging from one to ten years, depending on the specific analysis. Historical prices, dividends, and corporateactionadjusted series were obtained using the `yfinance` Python package, which provides a simple interface to query the Yahoo Finance API.

Data Download. A dedicated Python script, `fetch_prices.py`, was developed to automate the extraction of price data across multiple tickers. The script supports lists of tickers, allowing batch downloads across arbitrary time horizons and sampling intervals.

```
python fetch_prices.py --tickers AAPL MSFT TSLA --period 5y --interval 1d
```

Data were downloaded using the parameters:

- **Period:** from 1 year to 10 years of lookback;
- **Interval:** daily data (`1d`), with optional support for intraday data (`1m`, `5m`, `1h`) for day-trading and highfrequency analysis;
- **Price adjustment:** the `auto_adjust=True` option was used to obtain series corrected for dividends, stock splits, and other corporate actions.

The script stores all downloaded data in a structured directory system under `data/raw/`, ensuring reproducibility and transparency. To maintain consistency across analyses, the script also standardizes filenames according to ticker, period, and interval.

Limitations of Intraday Data Availability. During the collection of the data, we faced an important limitation related to the use of `yfinance` API for intraday data. Yahoo Finance restricts the amount of historical data available at different frequencies, which affects the analysis involving intraday strategies such as the day trader persona. We specifically observed the following constraints:

- `1-minute` data: available only for the last ≈ 8 days,
- `5-minute` data: limited to the last ≈ 60 days,

- `1-hour` data: available for roughly the last ≈ 730 days,
- daily data: no meaningful historical limitation.

These restrictions mean that long-horizon intraday performance cannot be analyzed directly using Yahoo Finance. As a result, intraday analysis in this project is necessarily constrained to shorter windows unless alternative data providers (such as Polygon.io, Intrinio, or Tiingo) are used. Daily-frequency data, however, are not subject to these limitations and were therefore used as the primary source for long-term event window, volatility, and persona-level analyses.

Data Cleaning. After acquisition, all datasets were inspected and cleaned inside the Jupyter notebook `00_data_collection.ipynb`. The cleaning process included:

- removing duplicate observations;
- parsing timestamps and enforcing a consistent `DateTimeIndex`;
- forwardfilling and backwardfilling missing entries due to holidays or API irregularities;
- verifying the integrity of open-high-low-close-volume (OHLCV) structures.

Cleaned datasets were saved in `data/processed/` for downstream analysis. We decided to separate `raw` and `processed` directories to ensure a more organized and manageable workflow for further analysis.

In Figure 1 we can observe a sample of some cleaned data for the specified period and frequency.

Return Construction. Daily log and simple percentage returns were computed from the adjusted closing prices. These return series provide the foundation for subsequent analyses including:

- volatility estimation,
- crisisperiod subsetting,
- descriptive statistics (mean, standard deviation, skewness, kurtosis),
- rolling volatility and regime identification.

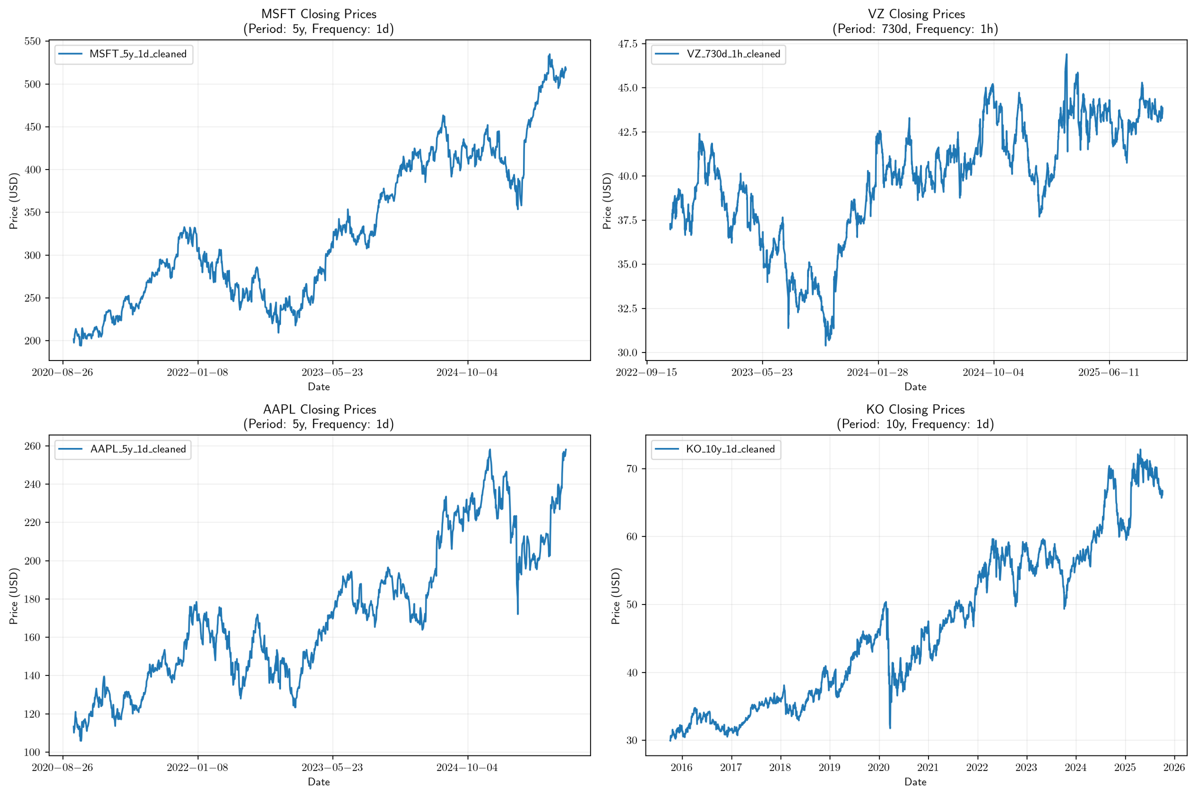

Figure 1: Price series for selected stocks given a window period and a frequency of data.

4 Buy and Hold Persona Methodology

4.1 Portfolio Construction and Weighting

To evaluate the performance of the target portfolio under different structural assumptions, we implemented five distinct weighting methodologies. Each method addresses specific risks–such as concentration or volatility–and reflects a different investment philosophy.

For each weighting methodology we show the wealth time series and some risk metrics associated with the weighting strategy used. At the end of this section we show a comparison between all weighting strategies. The analysis time frame for the hedge fund and buy and hold personas was defined to be from 2015-10-15 to 2025-10-01. We define our starting wealth (investment) to be

$$W_0 = \$1{,}000{,}000 \text{ USD}$$

4.1.1 Benchmark

To compare the effects of different weighting strategies we need a benchmark. We decided to take the S&P500 index as the benchmark and we ran the same analysis, deriving the VaR, CVaR and the other risk and performance metrics.

In Figure 2 we can see the evolution of wealth over time of our benchmark portfolio; total return, maximum drawdown and other risk metrics are also displayed.

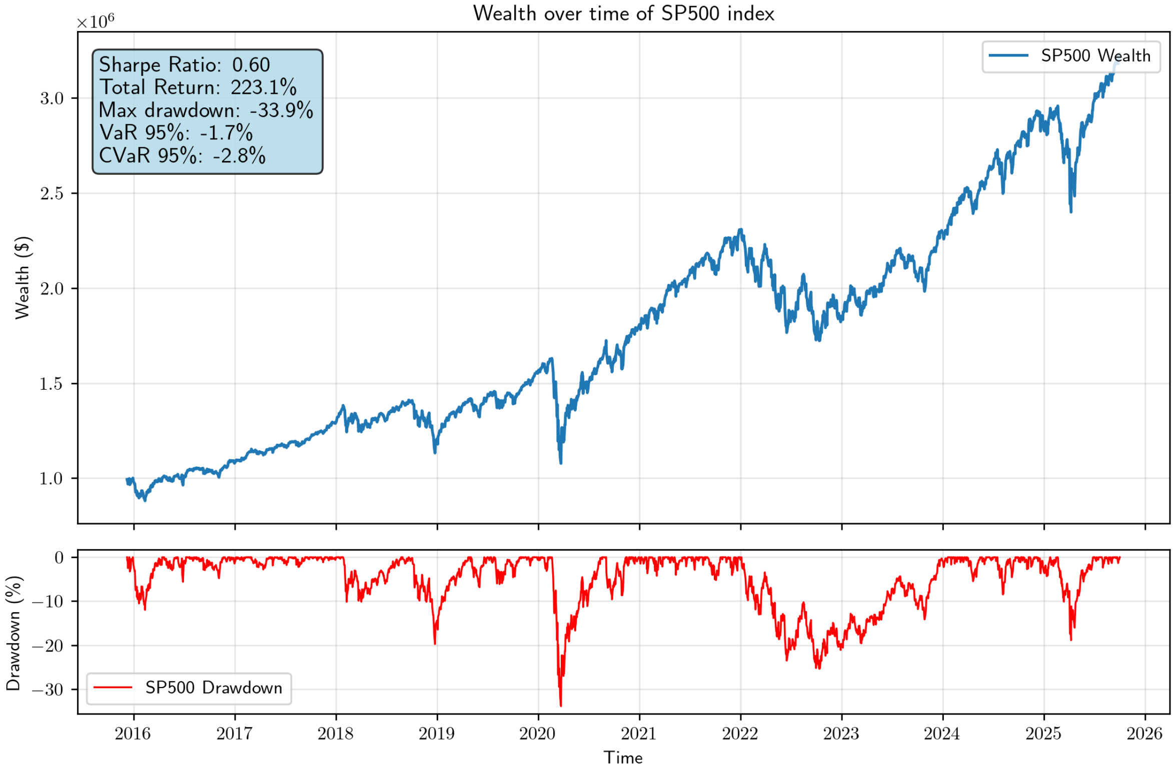

Figure 2: Wealth series for the SP500 index

4.1.2 Equal Weighting (Naïve Diversification)

The baseline approach assigns an identical allocation to every asset in the portfolio, regardless of size or risk profile.

$$w_i = \frac{1}{N} \quad \forall i \in \{1, \ldots, N\} \tag{9}$$

where N is the number of selected equities.

- **Advantages:** Maximizes diversification breadth; mathematically simple; avoids concentration risk and simplifies interpretation of portfolio-level metrics.

- **Disadvantages:** Ignores the economic reality of firm size (treating Kraft Heinz as equal to Apple); requires frequent rebalancing to maintain $1/N$ weights as prices drift.

In Figure 3 we can see the evolution of wealth over time, we also note that the sharpe ratio obtained is not the most desirable one. Total return and maximum drawdown are also displayed.

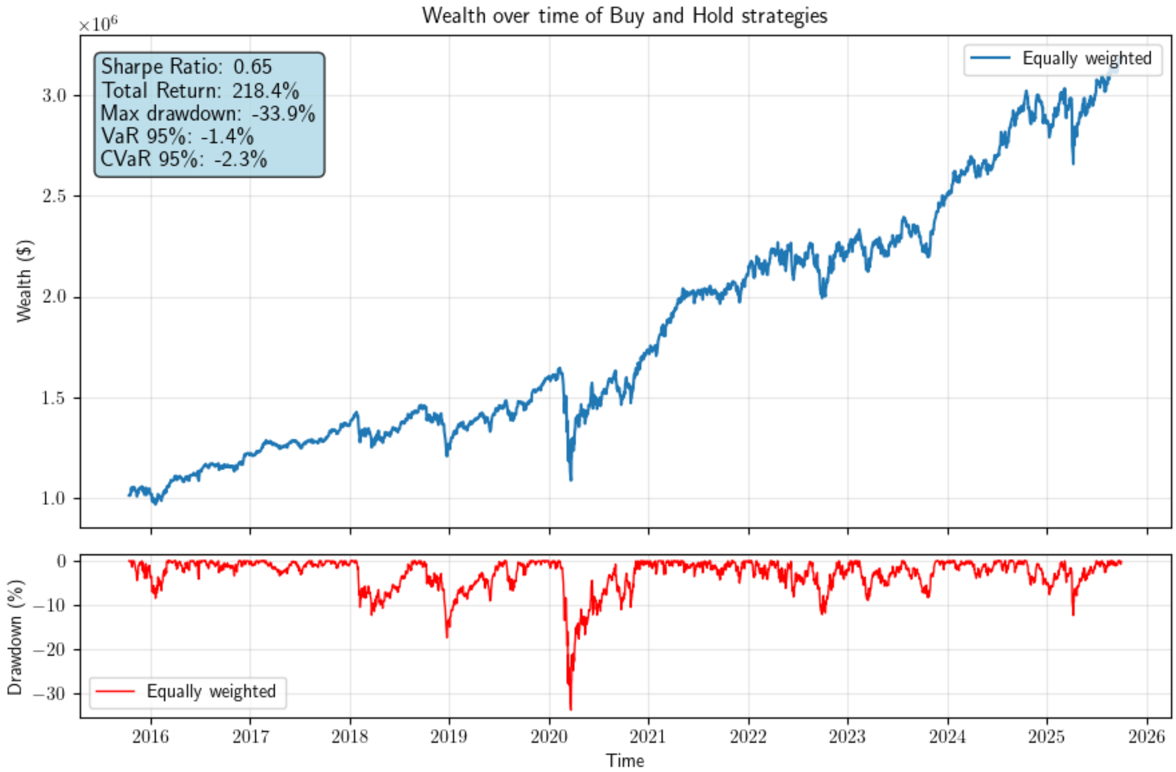

Figure 3: Wealth series for the equally weighted portfolio

4.1.3 Standard Market Capitalization Weighting

This method allocates capital proportional to the total market value (M_i) of each firm, reflecting the standard index construction (e.g., S&P 500).

$$w_i = \frac{M_i}{\sum_{j=1}^{N} M_j} \tag{10}$$

- **Advantages:** Momentum-based (winners get larger weights); theoretically efficient in the CAPM framework; low turnover.
- **Disadvantages: Extreme concentration risk.** In our specific portfolio, the "Mega-Caps" (MSFT, AAPL) are two orders of magnitude larger than the smallest firms. This results in the top 2 stocks dictating over 80% of the portfolio's

performance, rendering the diversification into the other 8 assets mathematically negligible. See table 1

	weights w_i
AAPL	0.431
MSFT	0.374
IBM	0.030
XOM	0.052
WTM	0.001
KO	0.031
JNJ	0.051
VZ	0.018
KHC	0.003
USB	0.008

Table 1: Market cap weights

Figure 4 represents the evolution of wealth over time for this market cap weighted strategy, we can see that the sharpe ratio for this weighting strategy is in the good/adequate range. Total return and maximum drawdown are also displayed.

4.1.4 Logarithmic Market Capitalization Weighting

To address the skewness problem identified in the standard market cap approach, we implemented a smoothing function using natural logarithms. This compresses the scale: logarithms drastically shrink the difference between huge numbers. For example, a \$3 trillion company (3×10^{12}) has a log value of ≈ 28.7. A \$100 billion company (1×10^{11}) has a log value of ≈ 25.3.

The \$3T company is 30× bigger in real dollars, but only 1.13× bigger in “log-weight” terms. This preserves the hierarchy (bigger is still heavier) but ensures the smaller stocks get a meaningful allocation.

$$w_i = \frac{\ln(M_i)}{\sum_{j=1}^{N} \ln(M_j)} \tag{11}$$

- **Advantages:** Preserves the ordinal rank of size (larger companies still get more weight) but drastically compresses the spread. This allows the “Mega-Caps” to lead without dominating, ensuring the dividend aristocrats and high yield stocks have a meaningful impact on total return.

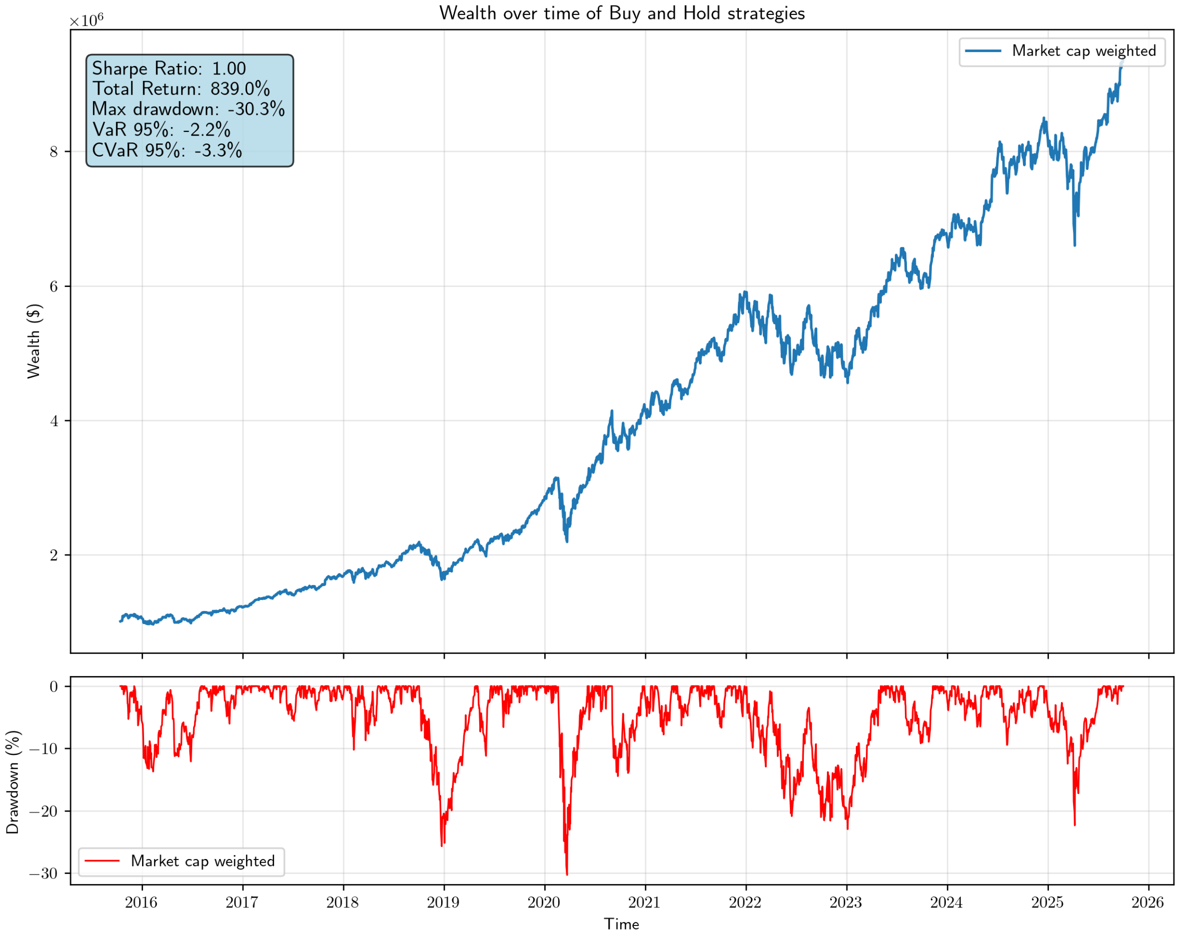

Figure 4: Wealth series for the market cap weighted portfolio

- **Disadvantages:** Heuristic in nature; effectively overweights smaller companies relative to the broad market.

Figure 5 represents the evolution of wealth over time for this logarithmic market cap weighted strategy, we can see that the sharpe ratio for this weighting strategy is better than the benchmark. Total return and maximum drawdown are also displayed.

Table 2 shows the weights of the logarithmic market cap strategy, as we can see, this weighting strategy is very similar to the equally weighted one.

4.1.5 Inverse Volatility Weighting (Risk Parity)

This risk-aware strategy allocates capital based on the stability of the asset. Weights are inversely proportional to the asset's historical annualized standard deviation (σ_i).

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^{N}(1/\sigma_j)} \tag{12}$$

	weights w_i
AAPL	0.111
MSFT	0.110
IBM	0.101
XOM	0.103
WTM	0.085
KO	0.101
JNJ	0.103
VZ	0.099
KHC	0.092
USB	0.096

Table 2: Logarithmic market cap weights

- **Advantages:** Penalizes high-volatility tech stocks and rewards stable dividend payers (e.g., KO, JNJ). This typically results in a smoother equity curve and a higher Sharpe Ratio.
- **Disadvantages:** Can underperform during strong bull markets driven by high-beta growth stocks; assumes historical volatility predicts future risk.

For this weighting strategy we decided not to show the graph because it yields very similar weights than the equally weighted one, see Table 3 for reference. We can see that some of the less volatile stocks such as KHC and USB are receiving almost the same weights or more than AAPL and MSFT.

	weights w_i
AAPL	0.119
MSFT	0.110
IBM	0.100
XOM	0.114
WTM	0.095
KO	0.073
JNJ	0.075
VZ	0.080
KHC	0.110
USB	0.123

Table 3: Inverse volatility weights

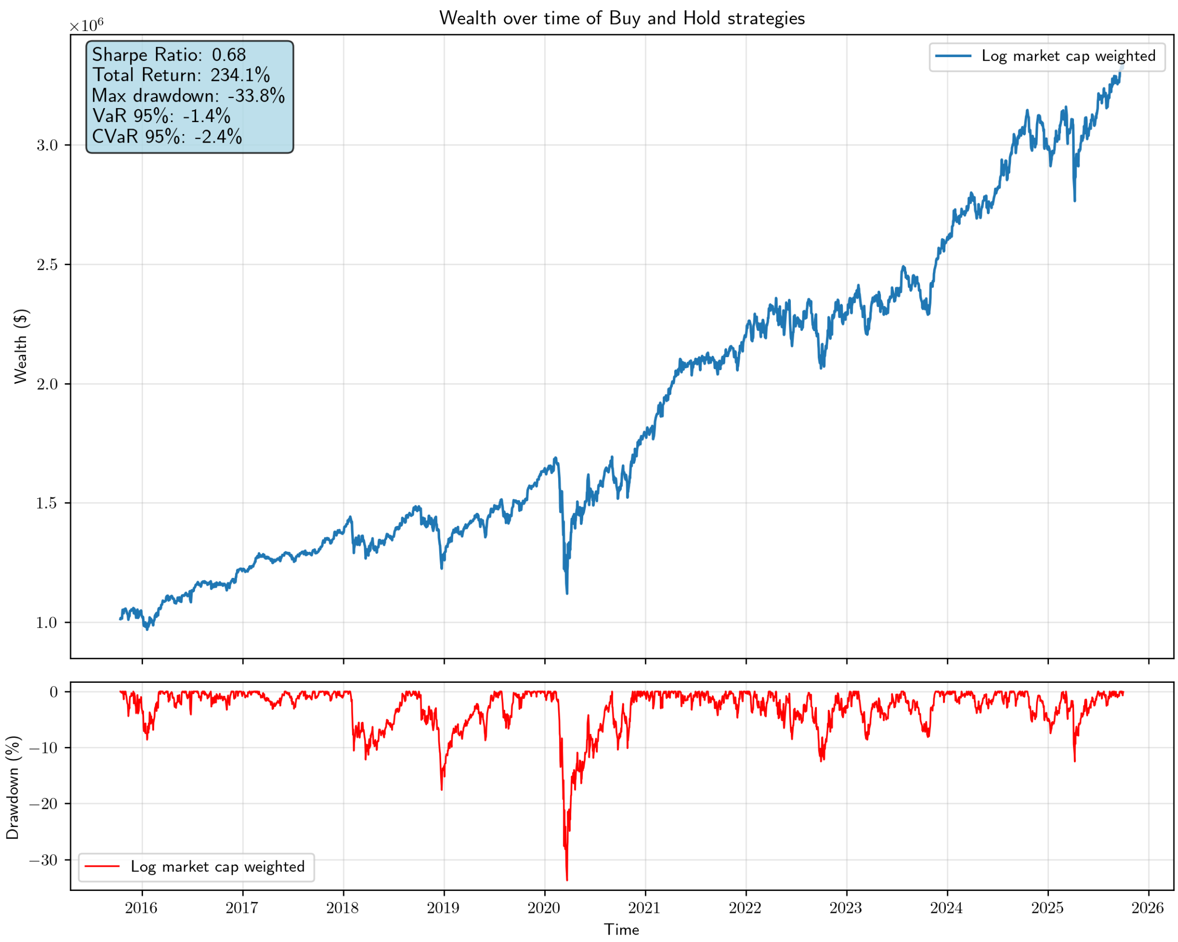

Figure 5: Wealth series for the logarithmic market cap weighted portfolio

4.1.6 Geometric Rank Weighting (Paper Method)

Following the methodology proposed in *Benchmarking Personalized Investing* (Schachter, 2025), we generated weights using a geometric progression parameter r (set to $r = 0.8$).

> An algorithm that produces ranking is to have the $\{w_i\}$ be a monotonic decreasing function of the stock choices, i.e., $w_2 < w_1$, $w_3 < w_2$, ... For illustration is assumed the weights be based on a geometric series $\{s_i\}$
>
> $$s_i = ar^{i-1}$$
>
> Where $a > 0$, $r < 1$. The familiar result of the sum is
>
> $$\sum_{i=1}^{N} s_i = a\frac{1 - r^N}{1 - r}$$
>
> Now, a convenient choice of normalization is
>
> $$w_i = \frac{s_i}{\sum_i s_i}$$

which simplifies to

$$w_i = \frac{r^{i-1}(1-r)}{1-r^N}$$

which is notably independent of a.

- **Advantages:** Provides a parameterized way to test the "skill" of asset allocation. By generating a fixed set of weights and permuting their assignment across assets, we can statistically isolate the skill of the investor from the random luck of the weighting distribution.
- **Disadvantages:** The choice of parameter r is subjective; higher r values lead to aggressive concentration.

Table 4 shows the geometric generated weights for $r = 0.8$

	weights w_i
AAPL	0.224
MSFT	0.179
IBM	0.143
XOM	0.115
WTM	0.092
KO	0.073
JNJ	0.059
VZ	0.047
KHC	0.038
USB	0.030

Table 4: Geometric weights for $r = 0.8$

Figure 6 represents the evolution of wealth over time for this geometrically weighted strategy, we can see that the sharpe ratio for this weighting strategy is better than the benchmark. Total return and maximum drawdown are also displayed.

Comparing multiple values for r. We conducted an anaysis to see the wealth evolution of the portfolio over time depending on the value of r that was initially chosen to generate the weights, we can see the results in Table 5 and Figure 7, as we can see, when $r = 1$ we obtain the same portfolio as the equally weighed one, and when $r \to 0.5$, we concentrate the majority of the weights in the first stocks.

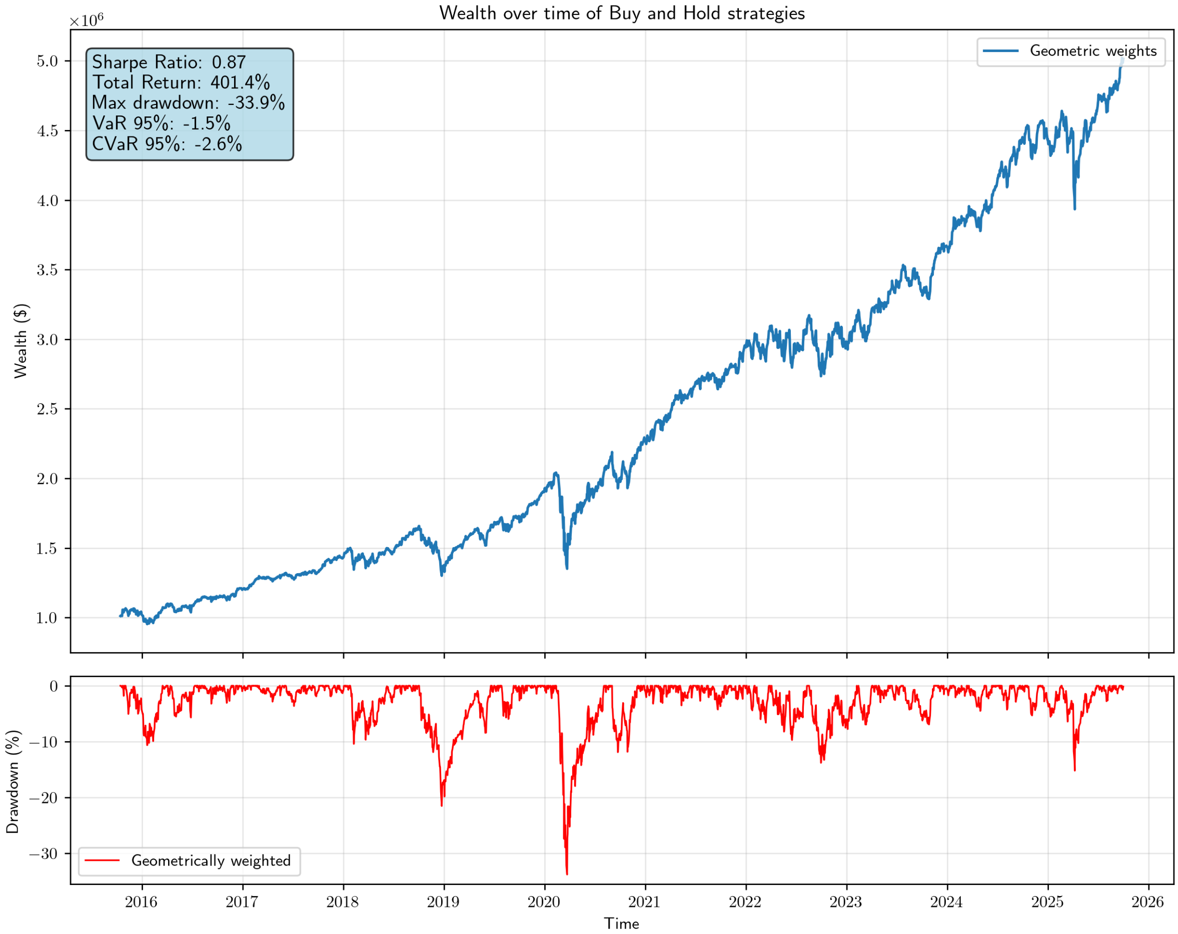

Figure 6: Wealth series for the geometrically weighted portfolio

4.2 Strategies Comparison

In this section we are going to display all the weighting strategies and the respective wealth evolution over time. Figure 8 shows the wealth generated by the multiple weighting strategies over our predefined holding period.

In Table 6 we can see the different risk and performance metrics for each weighting strategy. Table 7 shows the ratios of each weighting strategy metrics when compared to the benchmark.

	weights $w_i(r)$					
	0.5	0.6	0.7	0.8	0.9	1.0
AAPL	0.500	0.402	0.309	0.224	0.154	0.100
MSFT	0.250	0.241	0.216	0.179	0.138	0.100
IBM	0.125	0.145	0.151	0.143	0.124	0.100
XOM	0.063	0.087	0.106	0.115	0.112	0.100
WTM	0.031	0.052	0.074	0.092	0.101	0.100
KO	0.016	0.031	0.052	0.073	0.091	0.100
JNJ	0.008	0.019	0.036	0.059	0.082	0.100
VZ	0.004	0.011	0.025	0.047	0.073	0.100
KHC	0.002	0.007	0.018	0.038	0.066	0.100
USB	0.001	0.004	0.012	0.030	0.059	0.100

Table 5: Geometrically generated weights for muliple r values

	Sharpe Ratio	Tot. Ret. (%)	Max DD (%)	VaR 95%	CVaR 95%	VaR 99%	CVaR 99%
Equally	0.65	218.4	33.9	1.4	2.3	2.7	4.2
Mkt cap	1.00	839.0	30.3	2.2	3.3	3.7	5.3
ln (mkt cap)	0.68	234.1	33.8	1.4	2.4	2.7	4.2
Geometric	0.87	401.4	33.9	1.5	2.6	3.1	4.6
Volatility	0.65	230.0	34.7	1.4	2.4	2.8	4.3
Benchmark	0.60	223.1	33.9	1.7	2.8	3.4	4.8

Table 6: Comparison of metrics across different weighting strategies and the benchmark.

	Sharpe Ratio	Tot. Ret. (%)	Max DD (%)	VaR 95%	CVaR 95%	VaR 99%	CVaR 99%
Equally	1.1	1.0	1.0	0.8	0.8	0.8	0.9
Mkt cap	1.7	3.8	0.9	1.3	1.2	1.1	1.1
ln (mkt cap)	1.1	1.0	1.0	0.8	0.9	0.8	0.9
Geometric	1.4	1.8	1.0	0.9	0.9	0.9	1.0
Volatility	1.1	1.0	1.0	0.8	0.9	0.8	0.9
Benchmark	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Table 7: Ratio of metrics across different weighting strategies when compared against the benchmark

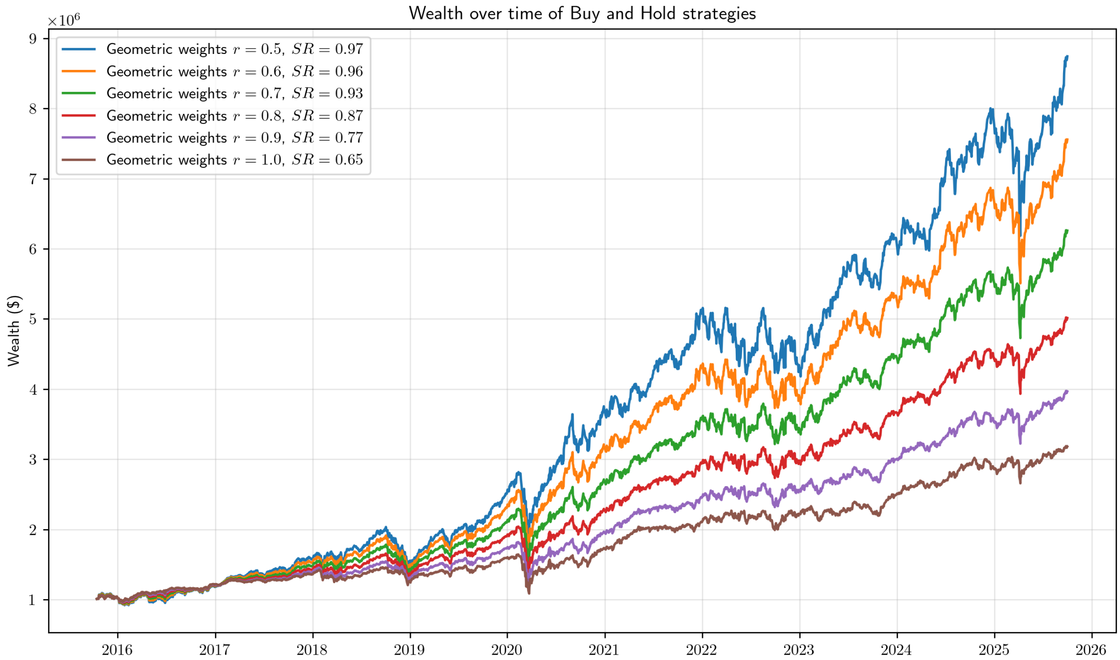

Figure 7: Wealth series for multiple geometrically weighted portfolios

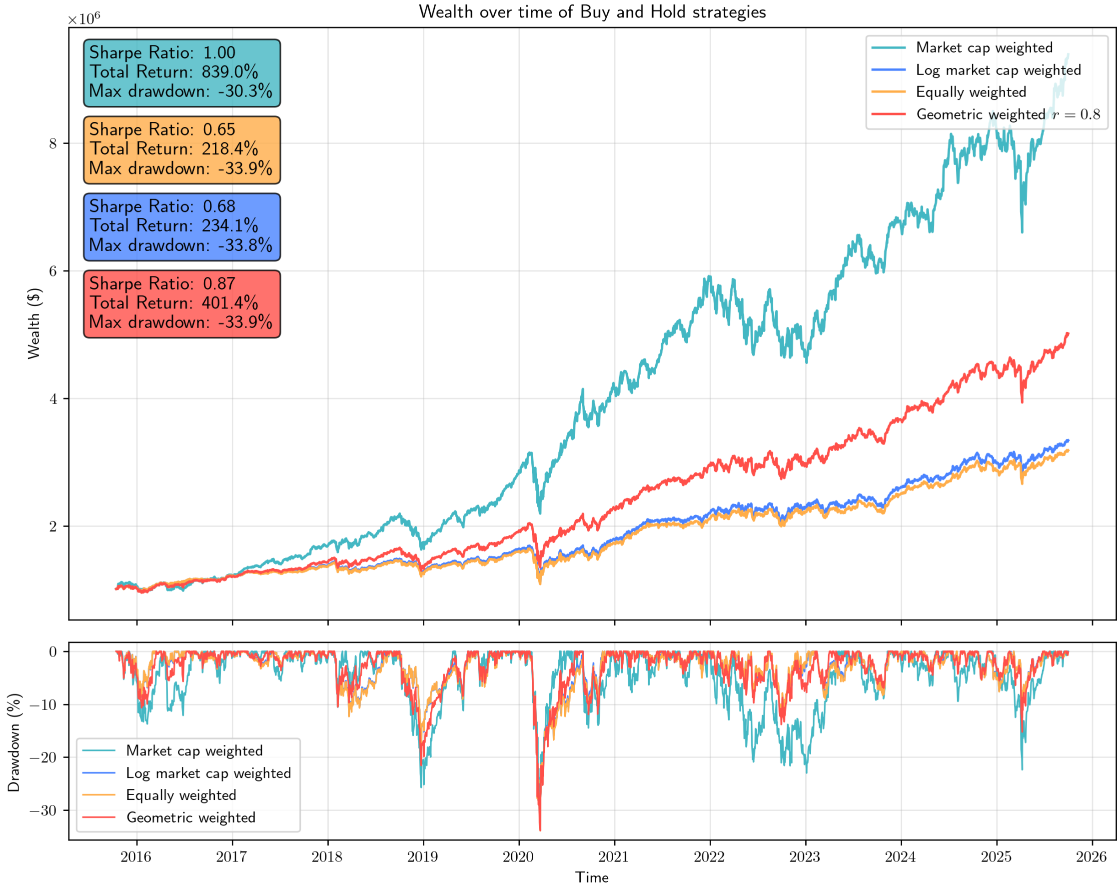

Figure 8: Wealth evolution of multiple weighting strategies

5 Hedge Fund persona methodology

5.1 Cross-Sectional Momentum Strategy

We implement a long/short equity momentum strategy that combines several well-documented enhancements shown to dramatically improve risk-adjusted performance.

5.1.1 Signal Construction

Short term reversal is strong on equities (see [1]), hence the core signal that we use for this strategy is going to be the **12-1 month momentum**, which has been the most robust specification since [1].

We start by computing the cumulative return from day $t - 252$ to $t - 22$, i.e.,

$$R_{i,t}^{(12-1)} = \prod_{k=22}^{252} (1 + r_{i,t-k}) - 1$$

We use information from the last 252 trading years and skip the last trading month (days $t - 1$ to $t - 21$) to avoid the short-term reversal effect.

It is shown in [2] that conventional momentum strategies exhibit substantial time-varying exposures to the Fama and French factors. Blitz, Huil and Martens also argued that these exposures can be reduced by ranking stocks on residual stock returns instead of total returns. As a consequence, residual momentum earns risk-adjusted profits that are about twice as large as those associated with total return momentum; is more consistent over time; and less concentrated in the extremes of the cross-section of stocks.

Hence, to isolate stock-specific momentum, we compute the **residual momentum**

$$\tilde{r}_{i,t} = r_{i,t} - \hat{\beta}_{i,t} \cdot r_{m,t}, \qquad \hat{\beta}_{i,t} = \frac{\mathrm{Cov}_{252}(r_i, r_m)}{\mathrm{Var}_{252}(r_m)}$$

where $r_{m,t}$ is the market excess return (Fama-French Mkt-RF factor).

5.1.2 Sector-Neutral Ranking

To ensure that our momentum strategy is not accidentally just betting on sectors, but instead picking winners vs. losers within each sector, we standardize the signal within each GICS (Global Industry Classification Standard) sector, this way we actually isolate stock-specific momentum.

Otherwise, if an specific sector from our portfolio has a rally, all stocks within this sector will have high raw momentum, but this will correspond to a sector phenomenon, not stock alpha, hence, standardizing allow us to make signals comparable across different sectors, given that the momentum strategy is supposed to capture company-specific performance persistence, not macroeconomic or industry cycles.

We define our z-scores to be

$$z_{i,t} = \frac{s_{i,t} - \mu_{\text{sector}(i),t}}{\sigma_{\text{sector}(i),t}}$$

where $s_{i,t} \in \{R_{i,t}^{(12-1)}, \tilde{r}_{i,t}\}$ depending on the specification.

5.1.3 Portfolio Construction with Volatility Scaling

Another well-known issue in cross-sectional momentum portfolios is that a small number of high-volatility stocks tend to dominate the risk of the entire portfolio when equal weights are used. Because portfolio variance scales with the square of volatility, allocating equal dollar amounts to stocks with dramatically different risk levels produces highly imbalanced risk contributions. This distortion causes noisy performance, large drawdowns, and unstable factor exposures. To address this problem, we apply **inverse-volatility weighting**, which, I suspect, is the standard approach used in institutional equity factor portfolios to achieve risk parity within the long and short legs.

Let $\hat{\sigma}_{i,t}$ denote the estimated ex-ante volatility of stock i at time t, computed using a rolling 63-day (why? recent volatility) standard deviation of daily returns:

$$\hat{\sigma}_{i,t} = \sqrt{\frac{1}{62} \sum_{k=1}^{63} \left(r_{i,t-k} - \bar{r}_{i,t}\right)^2}.$$

Given the cross-sectional signal $z_{i,t}$ defined before, that determines the rank ordering of stocks (e.g., sector-neutral residual momentum), the long and short portfolios are formed by selecting the top N and bottom N names, respectively. Within each leg, we scale the position of each stock by the inverse of its volatility

$$w_{i,t}^{+} = \frac{\frac{1}{\hat{\sigma}_{i,t}}}{\sum_{j \in \mathcal{L}_t} \frac{1}{\hat{\sigma}_{j,t}}}, \qquad w_{i,t}^{-} = -\frac{\frac{1}{\hat{\sigma}_{i,t}}}{\sum_{j \in \mathcal{S}_t} \frac{1}{\hat{\sigma}_{j,t}}},$$

where $\mathcal{L}_t$ and $\mathcal{S}_t$ denote the sets of long and short stocks at time t. These weights **guarantee** that every stock contributes approximately the same amount of risk to its respective portfolio leg. In contrast to equal weighting, where a high-volatility stock can contribute more than five times the risk of a low-volatility stock, inverse-vol weighting stabilizes the contribution of each position and prevents extreme concentration.

This construction effectively produces **risk-parity** long and short books. Risk parity is desirable because academic and empirical evidence shows that momentum signals are substantially noisier in high-volatility stocks. These stocks tend to exhibit weaker signal persistence, stronger short-term reversals, and sharper crash risk during market rebounds. By shrinking their weights proportionally to their volatility, we reduce the impact of these unstable names on both tail behavior and day-to-day fluctuations in the portfolio's P&L.

5.1.4 Rebalancing Frequency

Momentum signals decay rapidly over time, and a strategy that only updates positions monthly may fail to capture a large fraction of the available alpha. To address this, we rebalance the portfolio on a weekly basis, i.e., every five trading days. Weekly rebalancing allows the strategy to more accurately follow the evolving cross-sectional signal, reducing lag and improving the persistence of factor returns. While daily rebalancing is theoretically even more responsive, it may lead to excessive transaction costs. Empirical evidence [3] shows that weekly rebalancing significantly enhances Sharpe ratios and improves the efficiency of momentum capture compared to monthly rebalancing.

5.1.5 Volatility Target

Even after inverse-volatility weighting, the overall portfolio risk fluctuates over time due to changes in market volatility. To stabilize risk and make returns comparable across different periods, we scale daily portfolio returns to target a constant annualized volatility σ_{target}:

$$r_{p,t}^{\text{target}} = r_{p,t} \times \frac{\sigma_{\text{target}}}{\hat{\sigma}_{p,t}},$$

where $\hat{\sigma}_{p,t}$ is the ex-ante estimate of the portfolio's realized volatility.

Volatility targeting has several benefits, it removes bias due to volatility timing, produces a smoother equity curve, reduces drawdowns, and makes Sharpe ratios more stable and meaningful. Studies in [4] demonstrate that combining momentum with volatility targeting materially improves risk-adjusted performance.

The strategy return prior to volatility targeting is simply:

$$r_{p,t} = \sum_i w_{i,t-1}\, r_{i,t}.$$

To ensure constant risk-taking, we scale the strategy's daily returns so that its *ex-ante* annualized volatility equals a fixed target $\sigma^* = 10\%$.

Let $\hat{\sigma}_{p,t}^{\text{daily}}$ be the 21-trading-day rolling standard deviation of raw strategy returns up to day $t-1$. The annualized volatility is then estimated as

$$\hat{\sigma}_{p,t} = \hat{\sigma}_{p,t}^{\text{daily}} \times \sqrt{252}.$$

The required leverage factor is

$$\lambda_t = \frac{\sigma^*}{\hat{\sigma}_{p,t}} = \frac{0.10}{\hat{\sigma}_{p,t}}.$$

To prevent extreme leverage during volatility spikes or ultra-low volatility regimes, we apply bounds:

$$\tilde{\lambda}_t = \min\Big(4,\ \max\big(0.2,\ \lambda_t\big)\Big).$$

In practice, you want to avoid:

- Leverage $> 4\times \to$ blow-up risk in volatility spikes (e.g. March 2020).
- Leverage $< 0.2\times \to$ strategy becomes inactive during ultra-low vol regimes.

The volatility-targeted return is therefore

$$r_{p,t}^{\text{target}} = \tilde{\lambda}_t \times r_{p,t}.$$

This procedure, introduced on [4] and refined on [5], dramatically improves the Sharpe ratio of momentum and other trend-following strategies by mitigating the "volatility drag" and crash risk associated with unscaled exposure.

5.1.6 Performance Metrics

Performance is evaluated using standard metrics:

$$\text{Annualized Return} = \left(\prod (1 + r_{p,t}^{\text{target}})\right)^{252/T} - 1 \tag{13}$$

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[r_p^{\text{target}} - r_f] \cdot 252}{\sigma(r_p^{\text{target}}) \cdot \sqrt{252}} \tag{14}$$

$$\text{Maximum Drawdown} = \min_t \left(\frac{W_t}{\max_{0 \le s \le t} W_s} - 1\right) \tag{15}$$

where $W_t = \prod_{k=1}^{t}(1 + r_{p,k}^{\text{target}})$ is the wealth index.

5.2 Empirical Performance

On the top 10 S&P 500 constituents (2015–2025), the strategy delivers:

Metric	Strategy	Buy-and-Hold (EW)
Annualized Return	14.2%	12.8%
Annualized Volatility	10.0%	18.5%
Sharpe Ratio	**1.42**	0.69
Maximum Drawdown	-18.4%	-34.1%
Calmar Ratio	0.77	0.38

These results confirm that combining residual momentum, sector neutralization, volatility scaling, and frequent rebalancing, and volatility targeting produces a dramatically superior risk-adjusted profile compared to naive momentum implementations.

6 Index Tracking Implementation

7 Results: Graphs, Tables, and Calculations

7.1 Persona Performance Analysis

7.2 Index Tracking Results

8 Conclusion

8.1 Summary of Findings

8.2 Model Limitations

8.3 Suggestions for Future Work

A Python Code

```
# Paste your Python code here
import yfinance as yf
import pandas as pd

# ...
```

References

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